

# AIDS-1 Comparisons

## Simple Reflex Agent vs Model-Based Reflex Agent

| Factor                | Simple Reflex Agent      | Model-Based Reflex Agent       |
|-----------------------|--------------------------|--------------------------------|
| Internal state        | No internal state        | Maintains internal state       |
| World knowledge       | Only current percept     | Uses world model               |
| Decision basis        | Condition-action rules   | State + rules                  |
| Sensor history        | Ignores past             | Tracks unobserved aspects      |
| Handling partial info | Fails often              | Works better                   |
| Complexity            | Very low                 | Moderate                       |
| Example               | Thermostat, light switch | Autonomous car, robot explorer |

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## Uniform Cost Search vs Best First Search

| Factor             | Uniform Cost Search     | Best First Search       |
|--------------------|-------------------------|-------------------------|
| Search type        | Uninformed (blind)      | Informed (heuristic)    |
| Cost handling      | Uses path cost          | Uses heuristic estimate |
| Expansion priority | Lowest cumulative cost  | Lowest heuristic value  |
| Completeness       | Yes (finite costs)      | Not guaranteed          |
| Optimality         | Always optimal          | Not optimal             |
| Memory usage       | High (frontier storage) | High (open list)        |
| Example            | Shortest path in map    | Greedy route finder     |

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## Informed vs Uninformed Search + A\*

| Factor        | Informed Search  | Uninformed Search | A* Algorithm                   |
|---------------|------------------|-------------------|--------------------------------|
| Heuristic use | Uses heuristic   | No heuristic      | Yes, uses $f(n) = g(n) + h(n)$ |
| Efficiency    | More efficient   | Less efficient    | Very efficient                 |
| Optimality    | Not always       | BFS/UCSS optimal  | Yes (admissible heuristic)     |
| Completeness  | Often incomplete | BFS/DFS complete  | Yes                            |

| Factor      | Informed Search | Uninformed Search | A* Algorithm                |
|-------------|-----------------|-------------------|-----------------------------|
| Memory need | Moderate        | High (BFS)        | High                        |
| Example     | Greedy BFS, A*  | BFS, DFS, UCS     | Pathfinding in games        |
| Category    | Informed        | Uninformed        | Informed (heuristic search) |

## Unification vs Skolemization

| Factor                   | Unification                          | Skolemization                     |
|--------------------------|--------------------------------------|-----------------------------------|
| Purpose                  | Match logical terms                  | Eliminate existential quantifiers |
| Operation type           | Syntactic substitution               | Quantifier removal                |
| Output                   | Most general unifier                 | Skolem function/constant          |
| Preserves satisfiability | Yes                                  | Yes (but not equivalence)         |
| Used in                  | Resolution, Prolog                   | Converting to CNF                 |
| Replaces variables       | Variable to term                     | Existential to constant/function  |
| Example                  | $\text{unify}(P(x), P(a)) = \{x/a\}$ | $\exists x P(x) \rightarrow P(c)$ |

## Data Scientist vs Big Data Professional vs Data Analyst

| Factor        | Data Scientist      | Big Data Professional    | Data Analyst          |
|---------------|---------------------|--------------------------|-----------------------|
| Primary focus | Predictive modeling | Distributed systems      | Descriptive analytics |
| Tools         | Python, R, SQL      | Hadoop, Spark, Kafka     | Excel, SQL, Tableau   |
| Skill level   | Advanced ML/stats   | Cluster management       | Statistical basics    |
| Output        | Models, algorithms  | Data pipelines           | Reports, dashboards   |
| Data size     | Medium to large     | Very large (PB)          | Small to medium       |
| Question type | Why & predict       | How to store/process     | What happened         |
| Example role  | Churn prediction    | Spark streaming engineer | Sales dashboarding    |

## Data Science vs Business Analytics vs Big Data

| Factor       | Data Science             | Business Analytics   | Big Data               |
|--------------|--------------------------|----------------------|------------------------|
| Core goal    | Discover insights        | Drive decisions      | Handle volume          |
| Data volume  | Large                    | Small/medium         | Very large             |
| Techniques   | ML, DL, statistics       | Excel, SQL, BI tools | Distributed computing  |
| Output       | Predictive models        | KPIs, reports        | Infrastructure         |
| Example use  | Recommendation system    | Sales forecasting    | Clickstream processing |
| Time horizon | Long-term                | Short-term           | Infrastructure-long    |
| Users        | Engineers, statisticians | Managers, analysts   | Data engineers         |

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## Data Scientist vs Big Data Professional vs Data Analyst

| Factor        | Data Scientist      | Big Data Professional | Data Analyst        |
|---------------|---------------------|-----------------------|---------------------|
| Focus         | Advanced analytics  | Big data infra        | Business reporting  |
| Tools         | Jupyter, TensorFlow | HDFS, MapReduce       | Power BI, Excel     |
| ML skill      | High                | Low                   | Very low            |
| Engineering   | Moderate            | High                  | Low                 |
| Data cleaning | Yes                 | Pipeline level        | Basic               |
| Stakeholder   | Product team        | IT/DevOps             | Business units      |
| Example       | Fraud detection     | Optimize storage      | Monthly sales trend |

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## Barplot vs Histogram

| Factor            | Barplot          | Histogram           |
|-------------------|------------------|---------------------|
| Data type         | Categorical      | Continuous          |
| Bars separated    | Yes (gaps)       | No (touching)       |
| X-axis meaning    | Category names   | Binned intervals    |
| Ordering          | Reorderable      | Fixed numeric order |
| Height meaning    | Count or value   | Frequency density   |
| Example           | Car colors sales | Age distribution    |
| Cumulative option | No               | Yes (ogive)         |

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## Z-Test vs T-Test vs ANOVA

| Factor              | Z-Test                | T-Test                   | ANOVA                    |
|---------------------|-----------------------|--------------------------|--------------------------|
| Sample size         | Large ( $n > 30$ )    | Small ( $n < 30$ )       | Any, but balanced        |
| Population variance | Known                 | Unknown                  | Estimated within         |
| Groups compared     | 1 or 2                | 1 or 2                   | 3 or more                |
| Distribution        | Normal                | T-distribution           | F-distribution           |
| Test statistic      | Z-score               | T-statistic              | F-ratio                  |
| Example             | Population mean check | Medicine effect pre/post | Crop yield by fertilizer |
| Pairwise need       | No                    | Optional                 | Post-hoc tests           |

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## Univariate vs Multivariate EDA

| Factor               | Univariate EDA          | Multivariate EDA          |
|----------------------|-------------------------|---------------------------|
| Variables analyzed   | One variable only       | Two or more variables     |
| Goal                 | Understand distribution | Find relationships        |
| Graphical tools      | Histogram, boxplot      | Scatterplot, heatmap      |
| Statistical measures | Mean, variance          | Correlation, covariance   |
| Example question     | What is age spread?     | Age vs income relation    |
| Outlier detection    | Possible                | In joint space            |
| Dimensionality       | Low                     | High (curse of dimension) |

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## Business Intelligence vs Data Science

| Factor       | Business Intelligence  | Data Science              |
|--------------|------------------------|---------------------------|
| Focus        | Past & present         | Future prediction         |
| Data type    | Structured             | Unstructured + structured |
| Time horizon | Descriptive/diagnostic | Predictive/prescriptive   |
| Tools        | Tableau, Power BI      | Python, R, MLlib          |

| Factor  | Business Intelligence  | Data Science            |
|---------|------------------------|-------------------------|
| Users   | Managers, analysts     | Engineers, scientists   |
| Output  | Dashboards, reports    | Models, algorithms      |
| Example | Quarterly sales report | Customer churn forecast |

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## Linear Regression vs Logistic Regression

| Factor            | Linear Regression      | Logistic Regression      |
|-------------------|------------------------|--------------------------|
| Output type       | Continuous value       | Probability/class        |
| Equation          | $y = wx + b$           | $p = 1/(1+e^{-(wx+b)})$  |
| Loss function     | MSE                    | Log loss (cross-entropy) |
| Range             | $(-\infty, +\infty)$   | $(0, 1)$                 |
| Decision boundary | Not applicable         | Linear in log-odds       |
| Diagram           | Straight line fit      | S-shaped curve (sigmoid) |
| Example           | House price prediction | Email spam detection     |

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## SVM vs Logistic Regression

| Factor               | SVM                        | Logistic Regression |
|----------------------|----------------------------|---------------------|
| Core principle       | Maximize margin            | Maximize likelihood |
| Output               | Class label                | Probability         |
| Kernel trick         | Yes (nonlinear)            | No (linear only)    |
| Overfitting control  | Regularization parameter C | L1/L2 penalty       |
| Outlier sensitivity  | Low (support vectors)      | Moderate            |
| Performance high-dim | Excellent                  | Moderate            |
| Example              | Image classification       | Credit risk scoring |

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## Linear Regression vs Logistic Regression

| Factor             | Linear Regression         | Logistic Regression                   |
|--------------------|---------------------------|---------------------------------------|
| Dependent var      | Continuous                | Categorical                           |
| Formula            | $y = \beta_0 + \beta_1 x$ | $\log(p/(1-p)) = \beta_0 + \beta_1 x$ |
| Error distribution | Normal                    | Binomial                              |
| Estimation         | OLS                       | MLE                                   |
| Interpretation     | Slope change              | Odds ratio                            |
| Diagram            | Straight line             | Sigmoid curve                         |
| Example            | Temperature vs sales      | Pass/fail prediction                  |

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## Linear Regression vs Logistic Regression

| Factor              | Linear Regression | Logistic Regression |
|---------------------|-------------------|---------------------|
| Response type       | Quantitative      | Binary/categorical  |
| Model form          | Linear equation   | Logit function      |
| Residuals           | Homoscedastic     | Not applicable      |
| R-squared           | Used              | Pseudo R-squared    |
| Coefficient meaning | Unit change in y  | Log-odds change     |
| Diagram type        | Scatter + line    | Probability curve   |
| Example             | Fuel efficiency   | Tumor malignant?    |